Seattle-Area Underground Transit Tunnels: Infrastructure, Egress, and Security Disclosures

Executive Summary

This report analyzes publicly documented surface structures, emergency egress points, and ventilation shafts associated with underground transit tunnels in the Seattle metropolitan area. Driven by the National Fire Protection Association (NFPA) 130 standard, modern rail transit tunnels require strict architectural interventions to ensure passenger life safety, including cross-passage spacing of 800 feet and surface exit spacing of 2,500 feet.

Operating exclusively within publicly available data, this analysis evaluates the Downtown Seattle Transit Tunnel (DSTT), University and Northgate Links, Beacon Hill Tunnel, Downtown Bellevue Tunnel (East Link), and the planned West Seattle and Ballard Link Extensions (WSBLE). Furthermore, this report contextualizes these structures within the legal frameworks of the Washington Public Records Act (RCW 42.56) and federal Sensitive Security Information (SSI) regulations under 49 CFR Part 1520.

The findings indicate that while major tunnel portals, main station shafts, and overarching ventilation facilities are frequently identified in public planning documents, the granular coordinates of mid-tunnel cross passages and specific intrusion apparatuses are heavily redacted or withheld. This selective disclosure reflects statutory mandates designed to balance public infrastructure transparency with the necessity of protecting mass transit networks from malevolent disruption.

Regulatory Architecture: NFPA 130 and the Physics of Fire Evacuation

The architectural and engineering configurations of Seattle's underground transit network are heavily dictated by NFPA 130: *Standard for Fixed Guideway Transit and Passenger Rail Systems*. This standard establishes the minimum requirements for life safety, fire protection, and emergency egress in subterranean transit environments. Because transit tunnels are constrained environments with high occupant loads, the prevention of smoke inhalation and the rapid evacuation of passengers govern every aspect of tunnel design.

Under NFPA 130, strict evacuation timelines and spatial geometries are enforced. Stations must be designed to evacuate the maximum platform occupant load within four minutes, and the entire station environment to a safe point within six minutes. To accommodate this, the Code restricts the maximum travel distance to an exit in an enclosed station to 250 feet (76 meters). Egress capacities are calculated using precise metrics: platforms, corridors, and ramps are assumed to process 2.08 persons per inch of width per minute, with a travel speed of 124 feet per minute. Stairs, presenting a vertical obstacle, process occupants at 48 vertical feet per minute. When combined with the massive passenger loads expected during peak hours, these mathematical constraints dictate the massive widths required for subterranean concourses and

vertical exit shafts.
Within the active trainway tunnels, egress geometries require cross passages connecting twin-bore tunnels at maximum intervals of 800 feet (244 meters). This allows passengers to flee a fire incident in one tunnel by crossing through a fire-rated door into the parallel tube, which acts as a safe haven and an evacuation route. Research indicates that the 800-foot metric originated in the 1970s during the design of the Metropolitan Atlanta Rapid Transit Authority (MARTA) system, based on the maximum distance an average population could walk downstream of a fire before flashover occurred. Furthermore, if emergency exits to the surface are utilized instead of or alongside cross passages, the maximum permitted distance between these surface exits is 2,500 feet (762 meters).
Because NFPA 130 mandates that trains and tunnels operate in environments where smoke can be effectively managed, massive mechanical ventilation plants—frequently integrated into station headhouses or standalone surface shafts—are required. These fans must generate what the code defines as "critical velocity": the minimum steady-state velocity of ventilation airflow moving toward the fire within a tunnel that is required to prevent "backlayering" (the movement of smoke against the ventilation airflow). By preventing backlayering, the ventilation system maintains a tenable, smoke-free path for evacuating passengers and advancing emergency responders.

Downtown Seattle Transit Tunnel (DSTT)

Alignment and Operational History

The Downtown Seattle Transit Tunnel (DSTT) constitutes the historical spine of Seattle’s high-capacity transit network. First approved by the Metro Council in 1983 and opening in 1990 originally for dual-mode buses, the tunnel underwent a massive retrofit and reopened in 2007 to accommodate a joint-operations model featuring both hybrid buses and Link light rail trains. In March 2019, bus operations ceased entirely, dedicating the tunnel exclusively to the Link light rail system. In 2022, ownership of the DSTT was formally transferred from King County Metro to Sound Transit.
The tunnel spans approximately 1.3 miles beneath the commercial core of downtown Seattle. Its alignment begins at the northern portal near 9th Avenue and Pine Street (the site of the former Convention Place station, which was permanently closed in 2018 to accommodate the expansion of the Seattle Convention Center). From this northern stub, the tunnel travels west beneath Pine Street before executing a sharp 90-degree curve to run south beneath 3rd Avenue, eventually terminating at the southern portal near 5th Avenue South and South Jackson Street, where the International District/Chinatown Station is located.

Ventilation and Egress Structures

Due to its initial design for internal combustion and dual-mode hybrid buses, the DSTT possesses a robust, legacy ventilation infrastructure. When buses operated in the tunnel, they utilized a "hush mode" allowing them to run on stored electric power, but mechanical fans were still required to move massive volumes of air through the stations, serving vital life-safety and heat-dissipation roles. Even with the transition to pure light rail, human occupants and train propulsion systems produce significant heat and carbon dioxide that require continuous exhaust. While the movement of the trains acts as a piston to push air through the tunnel

naturally, mechanical fans remain a strict requirement for emergency fire suppression and worker safety.
The critical nature of this system was highlighted in 2019 when a severe ventilation failure at Pioneer Square Station required the total shutdown of the DSTT to satisfy fire code and worker safety mandates, forcing Sound Transit to run shuttle buses on surface streets.
A primary documented standalone ventilation structure for this alignment is the Pine Street Ventilation Facility. Designated as N00 in Sound Transit asset and numbering documents, this facility serves the northern stub of the tunnel where the DSTT transitions into the University Link segment. Additional venting occurs through street-level grates and mechanical rooms embedded within the structural footprints of Westlake, Symphony, and Pioneer Square stations, which are integrated into the existing civic architecture.

Publicly Documented Structures: DSTT

Structure	Type	Public location	Visible from street (Y/N)	Source	Confidence
N00 Pine Street Ventilation Facility	Vent / Fan Plant	Pine Street and 9th Avenue	Y	*Sound Transit Equipment and Facilities Numbering Standard*, Sound Transit, Aug 2021, p. 27, URL; *University Link Tunnel*, Wikipedia, URL	High: Explicitly named in ST asset logs and widely documented.
Pioneer Square Station Ventilation System	Vent	Pioneer Square Station	N (Internal)	*PSA: Light Rail not using Downtown Tunnel due to ventilation failure*, Reddit /r/Seattle, 2019, URL	High: Publicly documented due to operational failure closures.
Southbound Portal	Portal / Egress	5th Avenue S and S Jackson Street	Y	*Downtown Seattle Transit Tunnel*, Wikipedia, URL	High: Major geographic landmark for the 1 and 2 Lines.

University Link and Northgate Link Tunnels

Alignment and Operational History

The University Link (U-Link) and Northgate Link extensions represent the rapid northward continuation of the light rail network from the original DSTT spine. U-Link, which opened for revenue service on March 19, 2016, features 3.15 miles of twin-bored tunnels traveling from the

Pine Street Stub Tunnel in downtown Seattle to the University of Washington (UW) campus. The alignment proceeds northeast beneath Interstate 5, through the Capitol Hill Station (approximately 65 feet underground), and under the Montlake Cut to the UW Station (approximately 100 feet below ground).
The Northgate Link Extension, which opened for revenue service in 2021, adds 4.3 miles of transit alignment, extending the twin bores north from the UW Station beneath the university campus to deep underground stations in the U District (NE 45th Street and Brooklyn Avenue NE) and Roosevelt (NE 65th Street and 12th Avenue NE). Following Roosevelt, the alignment transitions to an elevated guideway via a portal near NE 94th Street toward the Northgate terminus.

Ventilation and Egress Structures

The engineering and environmental review process for U-Link involved significant public debate regarding the placement of ventilation shafts. Originally, a standalone ventilation shaft was proposed in the Montlake neighborhood to handle mid-tunnel smoke evacuation. However, this structure was removed from the final design in 2007 following intense community opposition and subsequent engineering modeling, which determined that the massive mechanical fan systems located at the UW and Capitol Hill stations could adequately manage emergency ventilation for the intervening tunnel span without a mid-point surface shaft.
To satisfy the NFPA 130 egress requirements along the 3.15-mile twin bores of U-Link without utilizing surface exits, the tunnels rely entirely on cross passages. According to Dr. Sauer & Partners, the engineering firm responsible for the soft-ground tunneling design, the U-Link twin-bored tunnels are connected by exactly 20 distinct cross passages. These allow passengers to evacuate laterally into the non-incident tunnel in the event of an emergency.
Further north, the Northgate Link Record of Decision and related environmental documents detail a critical combined tunnel portal and ventilation site situated between NE 58th and NE 60th Streets, positioned east of the Interstate 5 off-ramp. This structure, frequently referred to in construction updates as the NE 60th Street portal, facilitates the system's transition from subterranean to aerial alignment and houses vital ventilation components required for the northernmost tunnel segment.

Publicly Documented Structures: U-Link & Northgate

Structure	Type	Public location	Visible from street (Y/N)	Source	Confidence
U-Link Cross Passages (20 total)	Cross Passage	Subterranean between Downtown, Capitol Hill, and UW	N	*Soft Ground University Link*, Dr. Sauer & Partners, URL	High: Quantity is explicitly documented by the engineering firm; precise coordinates are protected.
NE 60th Street Portal and Vent Site	Vent / Portal	Between NE 58th and NE 60th Streets, east of the I-5	Y	*North Link Light Rail Transit Project Record of Decision*,	High: Explicitly detailed in the federal Record of Decision and

Structure	Type	Public location	Visible from street (Y/N)	Source	Confidence
		off-ramp		FTA/Sound Transit, p. 5, URL	construction updates.
Roosevelt Station Vent/Egress	Station Facility	NE 65th Street and 12th Avenue NE	Y	*Link Light Rail Progress Report*, Sound Transit, Q2 2016, URL	High: Standard underground station infrastructure location.

Beacon Hill Tunnel and Station

Alignment and Operational History

The Beacon Hill tunnel forms a critical 4,300-foot segment of the initial Central Link line (now the 1 Line), providing a high-capacity connection between the industrial flatlands of SODO and the Rainier Valley. Opened for revenue service in July 2009 after five years of intense construction, the twin running tunnels were excavated utilizing an Earth Pressure Balance (EPB) tunnel boring machine custom-built by Mitsubishi Heavy Industries. This resulted in twin tubes with an 18-foot 10-inch inner diameter, lined with a one-pass precast segmental lining. The tunnels reside approximately 160 feet beneath the crest of Beacon Hill, creating unique challenges for emergency egress.

Ventilation and Egress Structures

Because of the station's extreme depth (160 feet underground), traditional cut-and-cover construction from the surface was geotechnically and economically impossible. Instead, the station concourse and platforms were mined using the Sequential Excavation Method (SEM)—also known as the New Austrian Tunneling Method (NATM)—making it one of the largest soft-ground SEM tunnels ever constructed in the United States.

To provide ventilation and emergency egress compliant with NFPA 130 for a deep-mined station, two massive vertical shafts were constructed to connect the surface headhouse to the subterranean concourse. The Main Shaft features a massive 46-foot inner diameter and plunges 181 feet deep. It accommodates four high-speed elevators, an emergency staircase, ventilation shafts, and essential mechanical/electrical equipment. Because the elevators carry the primary egress load, they must be rated to function during fire emergencies, with their carrying capacity over 30 minutes factoring into the egress calculations.

Adjacent to the Main Shaft is the Ancillary Shaft. Boasting a 26-foot inner diameter, this shaft houses a secondary set of emergency staircases alongside supplementary ventilation shafts, ensuring redundancy in the event the primary shaft is compromised. At the bottom of these shafts, a 41-foot-wide Concourse Cross Adit provides passenger and emergency access to the platform tunnels. Dedicated ventilation tunnels branch off to provide airflow during normal operations and purge smoke during emergencies. The entire structure is designed as a "tanked" system, fully waterproofed to withstand biological and chemical deterioration from aggressive groundwater. The critical nature of these ventilation shafts was demonstrated in August 2025, when a ventilation fan issue at the station triggered a temporary halt of train service through the

tunnel to maintain compliance with fire code requirements.

Publicly Documented Structures: Beacon Hill

Structure	Type	Public location	Visible from street (Y/N)	Source	Confidence
Main Shaft (Elevators/Vent/Stairs)	Shaft	2702 Beacon Avenue South (Beacon Ave S and S Lander St)	Y (Headhouse)	*Benchmark for the Future: The largest SEM Soft Ground Tunnels in the US*, Dr. Sauer & Partners, 2013, p. 1-2, URL	High: Extensive engineering literature details the exact geometry and location.
Ancillary Shaft (Vent/Stairs)	Shaft	2702 Beacon Avenue South (Beacon Ave S and S Lander St)	Y (Headhouse)	*Benchmark for the Future: The largest SEM Soft Ground Tunnels in the US*, Dr. Sauer & Partners, 2013, p. 1-2, URL	High: Confirmed alongside the Main Shaft in engineering documents.

Downtown Bellevue Tunnel (East Link, 2 Line)

Alignment and Operational History

The East Link Extension (now designated the 2 Line) expands light rail service from Seattle, across the I-90 floating bridge, into the East King County municipalities of Mercer Island, Bellevue, and Redmond. A critical component of this corridor is the Downtown Bellevue Tunnel (designated as Contract E330), which allows the light rail to bypass dense surface traffic through the downtown Bellevue core. Rather than utilizing a tunnel boring machine, this segment was constructed using the Sequential Excavation Method (SEM), mining south to north directly beneath 110th Avenue NE. Systemwide revenue service linking this segment across Lake Washington to Seattle is anticipated in 2026.

Ventilation and Egress Structures

Public documentation indicates that the Bellevue Tunnel is outfitted with internal emergency ventilation fans that underwent extensive testing and commissioning as of late 2022 and early 2023. Due to the shorter length of this specific tunnel segment compared to the U-Link or Northgate bores, standalone mid-tunnel vertical ventilation shafts are not prominently documented in the public domain. Instead, it is highly likely that ventilation and smoke evacuation are managed at the portal ends and via the mechanical infrastructure of the adjacent cut-and-cover station boxes.
Regarding physical security at the egress points, public records from the City of Bellevue note

that unlike some older systems, the East Main portal of the Bellevue tunnel will not feature physical gates or doors blocking the trackway. Instead, advanced intrusion detectors will be installed that immediately alert the operations control center to any unauthorized pedestrian access.

Publicly Documented Structures: Downtown Bellevue

Structure	Type	Public location	Visible from street (Y/N)	Source	Confidence
Downtown Bellevue Tunnel Alignment	Tunnel / Vents	Subterranean along 110th Ave NE	N	*Agency Progress Report*, Sound Transit, Dec 2023, p. 5, URL	High: Alignment is verified; internal vent fans are noted in progress reports.
East Main Portal	Portal	Near East Main Station	Y	*East Link Light Rail Pedestrian & Bicycle Access*, City of Bellevue, July 2015, URL	High: City documentation confirms portal alignment and intrusion detection systems.

Tunnels Under Construction or Planned: WSBLE and DSTT2

Alignment and Operational History

The West Seattle and Ballard Link Extensions (WSBLE) represent a massive, multi-billion-dollar expansion of the regional transit network, slated for phased completion in the late 2030s and early 2040s. The Ballard segment includes 7.1 miles of light rail, running from downtown Seattle to Ballard via a newly constructed, rail-only tunnel through the urban core. This project necessitates the construction of a second downtown transit tunnel (DSTT2), designed to relieve capacity constraints on the original DSTT by separating the lines. The new tunnel will navigate deep subterranean environments through South Lake Union, Seattle Center, and a proposed secondary Midtown station, before crossing beneath Salmon Bay to reach Ballard.

Ventilation and Egress Structures

Because WSBLE remains in the Environmental Impact Statement (EIS) and preliminary engineering phases, specific land parcels for ancillary structures are subject to ongoing refinement. However, Draft EIS documents identify the critical infrastructure footprints required for these deep tunnel alignments. For example, documents reference the requirement for massive tunnel portals and shaft excavations, specifically noting a proposed vent shaft structure situated on the northwest corner of a station block in the Downtown/Midtown/Harrison segment. Due to the extreme depths proposed for the DT-1 Midtown Station—which must tunnel *under*

the existing DSTT to reach its alignment—sophisticated ventilation and emergency egress shafts will be mandatory to comply with NFPA 130. The increased depth heavily drives the cost and complexity of the shaft configurations, as vertical egress travel times increase exponentially. The debate over the necessity of DSTT2 has generated significant civic friction, with some analysts arguing that the second tunnel builds double the capacity Sound Transit actually needs, and that the physical addition of new ventilation shafts in the middle of dense urban blocks creates unnecessary engineering and financial burdens.

Publicly Documented Structures: WSBLE

Structure	Type	Public location	Visible from street (Y/N)	Source	Confidence
Midtown / Downtown Vent Shafts	Vent / Shaft	Downtown Seattle (Exact parcel pending final EIS)	TBD	*WSBLE Draft EIS: Alternatives Evaluation*, Sound Transit, Jan 2022, p. 6-2, URL	Low: Structures are mandated by code, but exact parcels remain under environmental review.

Contextual Tunnels (Non-Transit)

For civic and geographical context, two non-transit underground structures hold prominence in Seattle's infrastructure topography and are occasionally conflated with the transit network:

- **SR 99 Tunnel (WSDOT):** This massive bored tunnel replaced the seismically vulnerable Alaskan Way Viaduct. Owned by the Washington State Department of Transportation (WSDOT), it accommodates double-deck highway traffic. Life safety, vehicular exhaust, and emergency egress are managed by massive, highly visible ventilation buildings located at the north and south portals. These multi-story structures house mechanical, electrical, and air-scrubbing equipment capable of purging thousands of cubic feet of exhaust per minute.
- **Seattle Underground (Pioneer Square):** This is a network of subterranean passageways and basements in the historic Pioneer Square neighborhood, created when the city rebuilt its streets one to two stories higher following the Great Seattle Fire of 1889. It serves as a historical and tourist site and is entirely disconnected from modern rail transit infrastructure.

Information Security and the Legal Framework of Civic Mapping

The publication, mapping, and extraction of infrastructure data for transit tunnels exist at the nexus of public transparency and homeland security. The exact coordinates, blueprints, and security mechanisms of these structures are heavily regulated by state and federal laws designed to thwart terrorism, sabotage, and unauthorized access.

Washington Public Records Act (RCW 42.56)

The Washington Public Records Act (PRA) establishes a broad mandate for transparency, operating under the presumption that all government records are open for public inspection. However, this presumption is counterbalanced by specific, categorical exemptions intended to protect critical infrastructure and individual privacy.

Under RCW 42.56.420, records relating to security and terrorism are explicitly exempt from public disclosure. Specifically, this statute protects:

- Specific and unique vulnerability assessments or specific and unique response or deployment plans.
- Information regarding the public and private infrastructure and security of computer and telecommunications networks.
- Security risk assessments, and security test results to the extent they identify specific system vulnerabilities.

The PRA requires exemptions to be narrowly construed, meaning an agency cannot blindly withhold an entire document if only a portion is sensitive; it must redact the exempt material and provide the remainder. However, when requested documents contain engineering blueprints of emergency egress doors, the exact GPS coordinates of mid-tunnel cross-passages, or the operational logic of SCADA (Supervisory Control and Data Acquisition) networks, transit agencies will universally apply the RCW 42.56.420 exemption to withhold those details. While agencies freely release high-level Environmental Impact Statements (which identify intersections for vent shafts to satisfy zoning laws), they will heavily redact the granular engineering data.

Agencies must act in good faith when responding; under RCW 42.56.550, if an agency erroneously withholds a public record, a prevailing requestor can be awarded attorney's fees, costs, and daily monetary penalties ranging from $5 to $100 per day that the record was unlawfully denied. Conversely, if an agency releases protected security information, it faces severe federal repercussions.

Federal Sensitive Security Information (49 CFR Part 1520)

Because Sound Transit and King County Metro utilize federal funding and operate under the oversight of the Federal Transit Administration (FTA) and the Transportation Security Administration (TSA), they are legally bound by Sensitive Security Information (SSI) regulations codified in 49 CFR Part 1520 (and corresponding rail security directives in 49 CFR Part 1580). SSI is unclassified information that, if publicly distributed, would be detrimental to transportation security. Under 49 CFR 1520.5, covered information includes threat assessments, specific security measures, and critical infrastructure details. Access to SSI is strictly limited to "covered persons" (such as transit employees, TSA inspectors, and designated local officials) who possess a demonstrated "need to know".

Transit agencies are federally forbidden from releasing SSI to the general public, overriding state laws. If a state-level PRA request encompasses SSI, the agency must withhold or redact the protected data. Furthermore, any documents containing SSI must carry a specific warning label dictating that unauthorized release may result in civil penalty or other action.

Navigating Public Records Requests

To file a records request with Sound Transit or King County Metro, the public must utilize their respective online portals. Both agencies (along with many regional municipalities, such as the City of Seattle and Mercer Island) currently employ commercial public records software

platforms like GovQA or NextRequest.

- **What to ask for:** Requests must be specific and identifiable. Asking for "all environmental review documents, scoping reports, and Draft EIS exhibits related to the NE 60th St portal" will yield voluminous, heavily redacted results. Conversely, asking for "blueprint schematics of tunnel intrusion alarm systems" will be summarily denied under RCW 42.56.420 and federal SSI rules.
- **Response Times:** Under RCW 42.56.520, the agency is legally required to provide a preliminary response within five business days. Within that window, the agency must either provide the record, provide an internet link to the record, acknowledge the request and give a reasonable estimate of time to produce the records, ask for clarification, or formally deny the request citing the specific statutory exemption. Due to the complexity and volume of transit-related requests, large inquiries are frequently delivered in digital installments over a period of weeks or months.

Legal Risks of Publishing Infrastructure Maps

Publishing a civic interest map detailing transit infrastructure carries inherent legal risks, primarily centered on civil liability and the unintentional dissemination of protected security information. While aggregating publicly available, pre-existing data (such as DEIS reports, WSDOT records, and news articles) is fundamentally protected under the First Amendment, combining this open-source intelligence with site-specific vulnerability assessments, or publishing unlawfully obtained engineering blueprints, could invite intense federal scrutiny from the TSA or the Department of Homeland Security.

Recommended Disclaimers: Any historical or civic map published based on this data must include strong, prominent disclaimers to limit liability and deter illicit activity. The map should explicitly state:

1. *All information contained within this map is sourced strictly from pre-existing, publicly accessible government reports, environmental impact statements, and news publications.*
2. *No independent security vulnerability analyses have been conducted, and no Sensitive Security Information (SSI) is depicted.*
3. *This map is provided for historical, architectural, and civic educational purposes only. Unauthorized entry into transit facilities, tunnels, or ventilation shafts is illegal, highly dangerous, and subject to severe prosecution.*

Gaps in the Public Record

The exact parameters, spatial coordinates, and operational mechanics of Seattle's underground transit network will remain partially obscured due to the aforementioned legal exemptions. The public record contains definitive gaps intentionally designed to protect the system. Notable omissions include:

- **Cross-Passage Coordinates:** While engineering firms publicly confirm the existence of 20 cross passages on the U-Link alignment to demonstrate NFPA 130 compliance, their exact depths, geographic coordinates, and the specifications of their fire-rated doors are entirely withheld.
- **Intrusion Detection Systems:** Mentions of "intrusion detectors" at the East Main Bellevue portal are public in community planning packets, but the specific technology utilized (e.g., infrared beams, microwave sensors, LIDAR) and the locations of the blind

spots or control panels are completely redacted.

- **Detailed Draft EIS Parcels:** The WSBLE project explicitly acknowledges the need for massive ventilation shafts in the Midtown and Downtown segments. However, until the Final EIS and Record of Decision are published, the precise parcels subject to eminent domain remain fluid, legally undefined, and absent from the public mapping layers.
- **SCADA Network Infrastructure:** The locations of the Supervisory Control and Data Acquisition (SCADA) nodes that control the tunnel ventilation fans, traction power substations, and emergency lighting are protected under RCW 42.56.420 as critical telecommunications infrastructure.

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